

Loan Decision Analysis & Predictive Model Building

Khoa Pham

Data Quality

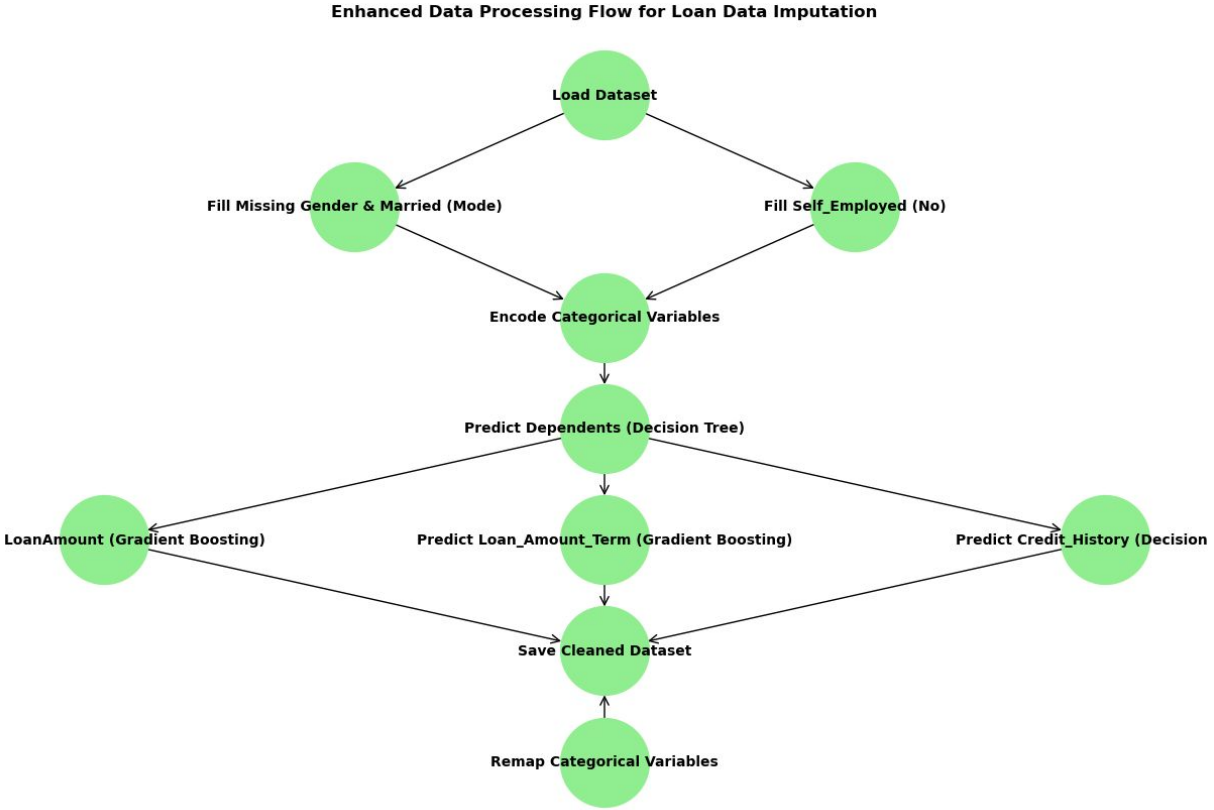
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 614 entries, 0 to 613
Data columns (total 13 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Loan_ID            614 non-null    object
 1   Gender             601 non-null    object
 2   Married            611 non-null    object
 3   Dependents         599 non-null    object
 4   Education          614 non-null    object
 5   Self_Employed      582 non-null    object
 6   ApplicantIncome    614 non-null    int64
 7   CoapplicantIncome  614 non-null    float64
 8   LoanAmount         592 non-null    float64
 9   Loan_Amount_Term   600 non-null    float64
 10  Credit_History     564 non-null    float64
 11  Property_Area      614 non-null    object
 12  Loan_Status        614 non-null    object
dtypes: float64(4), int64(1), object(8)
memory usage: 62.5+ KB
```

```
df.isnull().sum()
```

```
Loan_ID              0
Gender              13
Married              3
Dependents          15
Education            0
Self_Employed       32
ApplicantIncome      0
CoapplicantIncome    0
LoanAmount          22
Loan_Amount_Term    14
Credit_History      50
Property_Area        0
Loan_Status          0
dtype: int64
```

Data Processing




```
df.info()
```

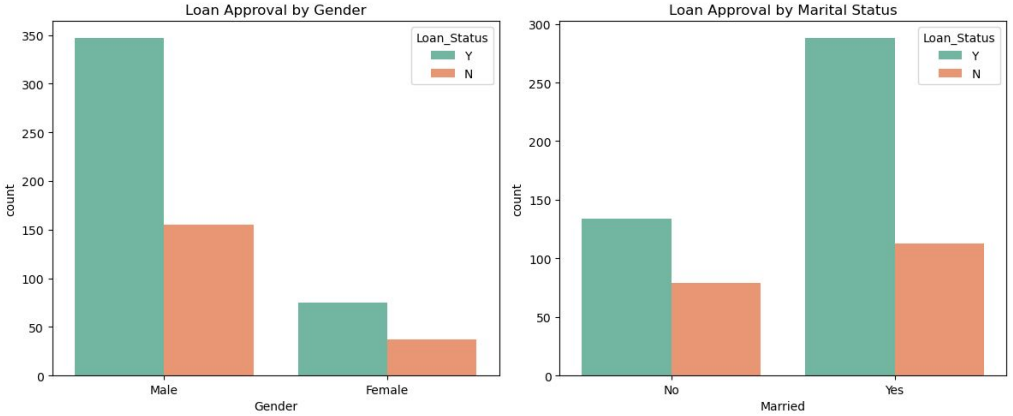
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 8   LoanAmount         614 non-null    float64
 9   Loan_Amount_Term   614 non-null    float64
 10  Credit_History     614 non-null    float64
 11  Property_Area      614 non-null    object 
 12  Loan_Status        614 non-null    object 
dtypes: float64(4), int64(5), object(4)
memory usage: 62.5+ KB
```

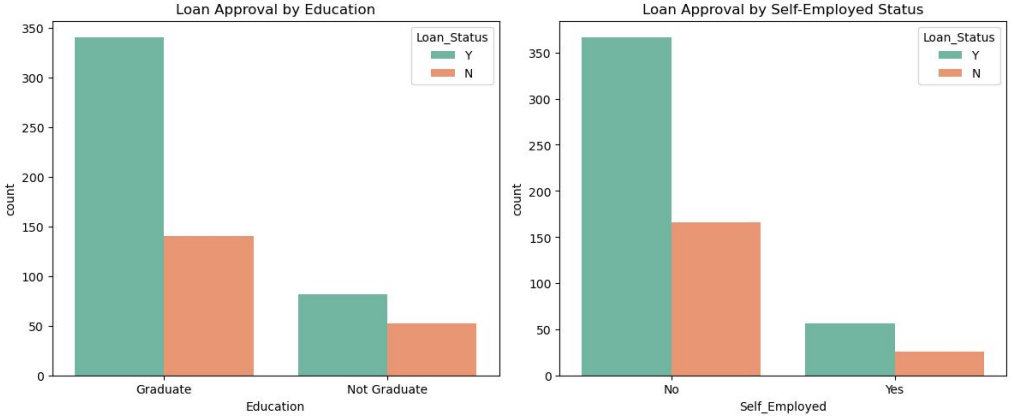
Loan Approval Rate Analysis



About 68% of applicants are approved for loans, while 32% are denied.

Loan Approval Rate Analysis





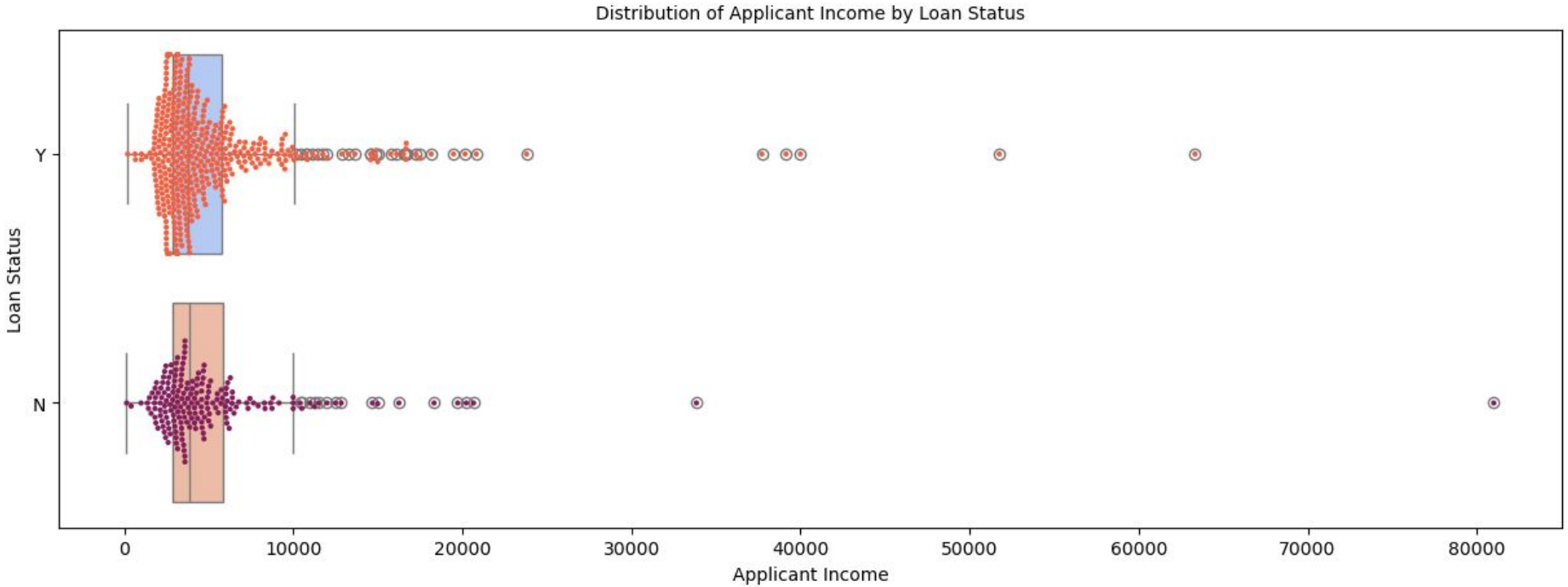
✔ Gender: Loan approval rates are slightly higher for male applicants, but the difference is small.

✔ Marital Status: Married individuals have a higher approval rate compared to unmarried applicants.

✔ Education: Graduates receive more loan approvals than non-graduates.

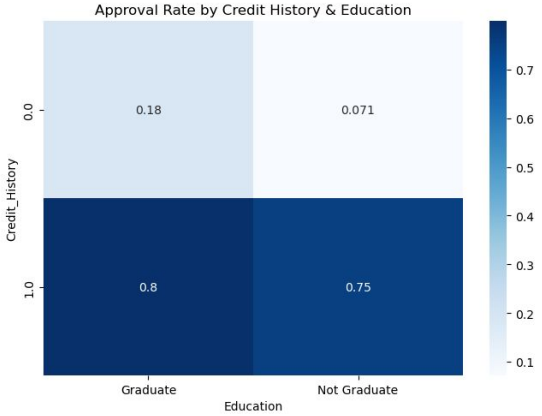
✔ Self-Employment: Self-employed applicants have a slightly lower approval rate, indicating income stability might be a factor.

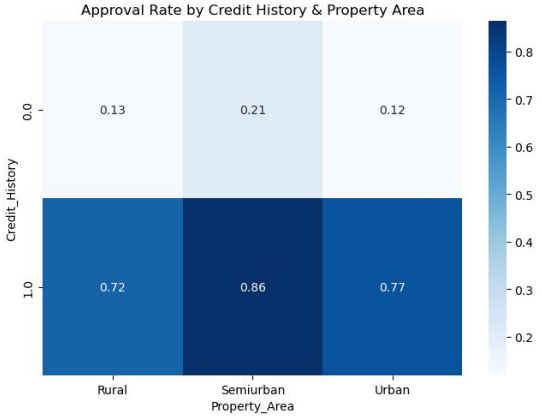
Loan Approval Rate Analysis

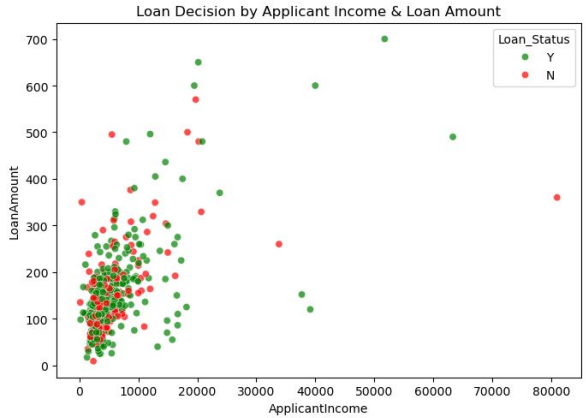


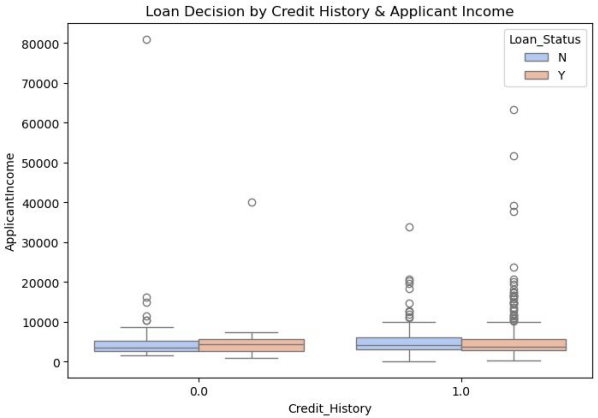
Applicant Income: Higher incomes do not guarantee approval, suggesting other factors (like credit history) are involved.

Loan Approval Rate Analysis









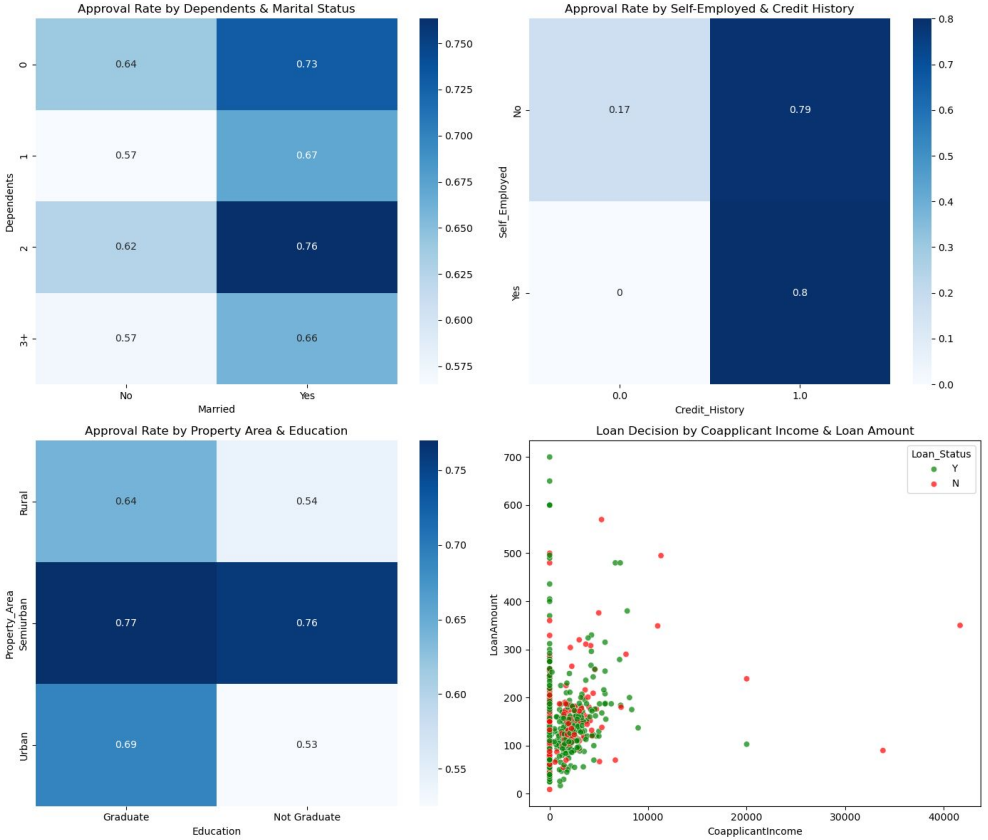
✔ Credit History is the #1 factor influencing loan approval.

✔ Property location affects approval likelihood, with semi-urban areas being more favorable.

✔ Loan Amount is a major risk factor, with higher amounts facing more rejections.

✔ Income alone does not guarantee approval—credit-history matters more.

Loan Approval Rate Analysis



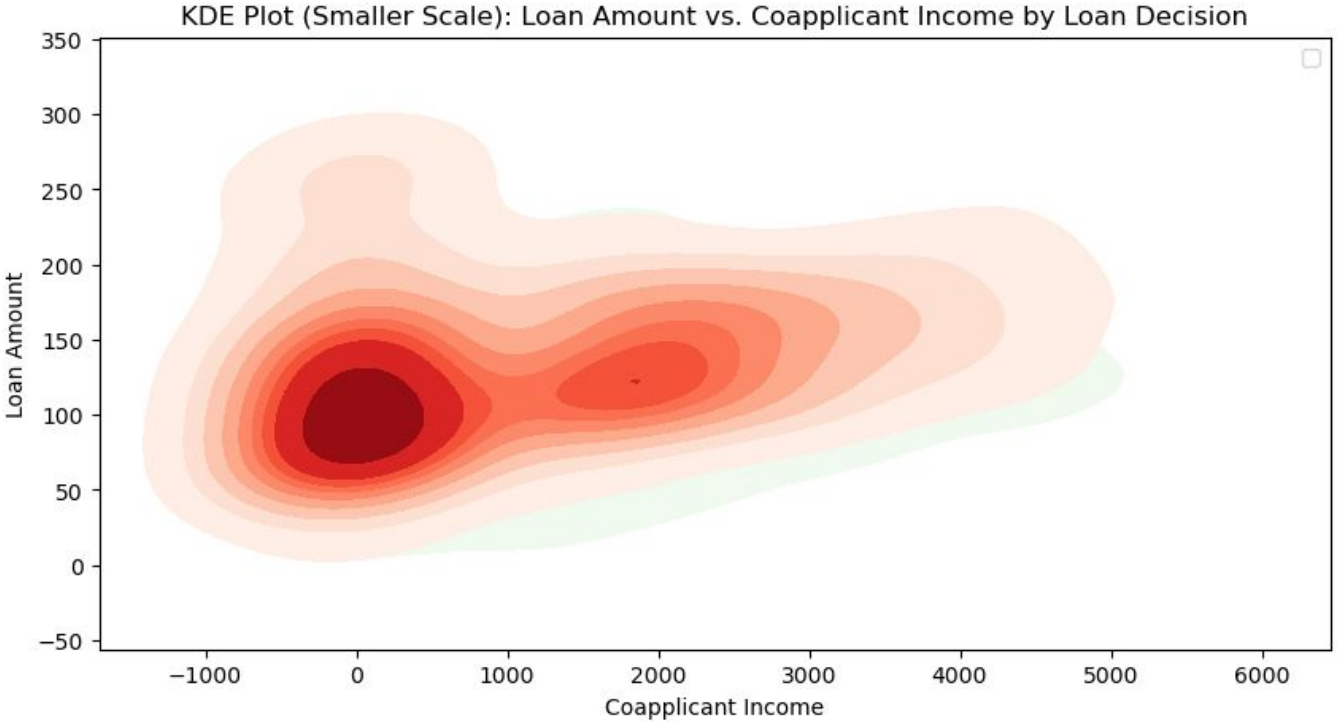
✔ Married applicants with dependents face higher rejection rates.

✔ Self-employed individuals must have a strong credit history for approval.

✔ Education matters, especially in rural areas where non-graduates struggle.

✔ Having a co applicant significantly improves approval chances for large loans.

Loan Approval Rate Analysis

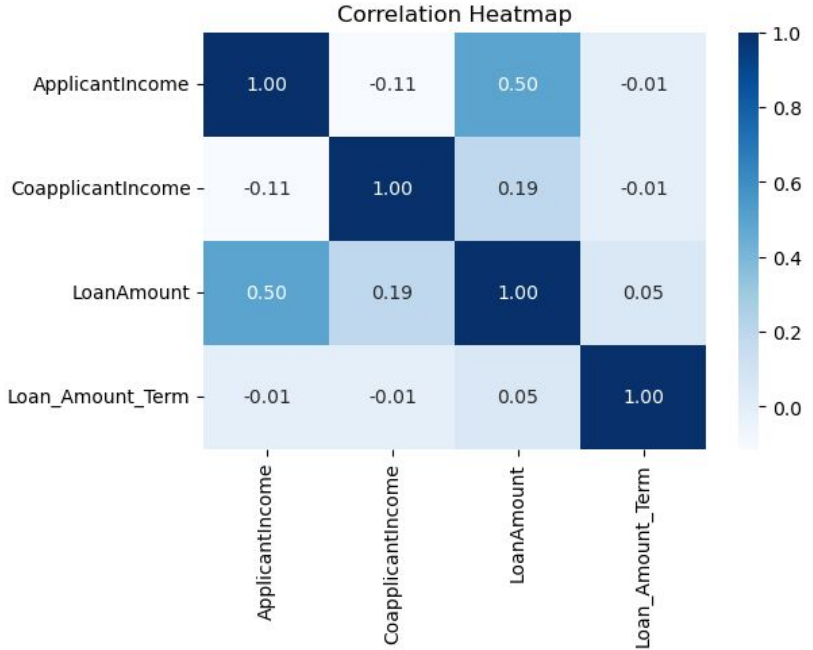


✔ Approved and rejected loans exhibit overlapping distributions, but rejected loans appear denser in certain regions.

✔ Loan approvals seem more concentrated at moderate co-applicant income levels and mid-range loan amounts.

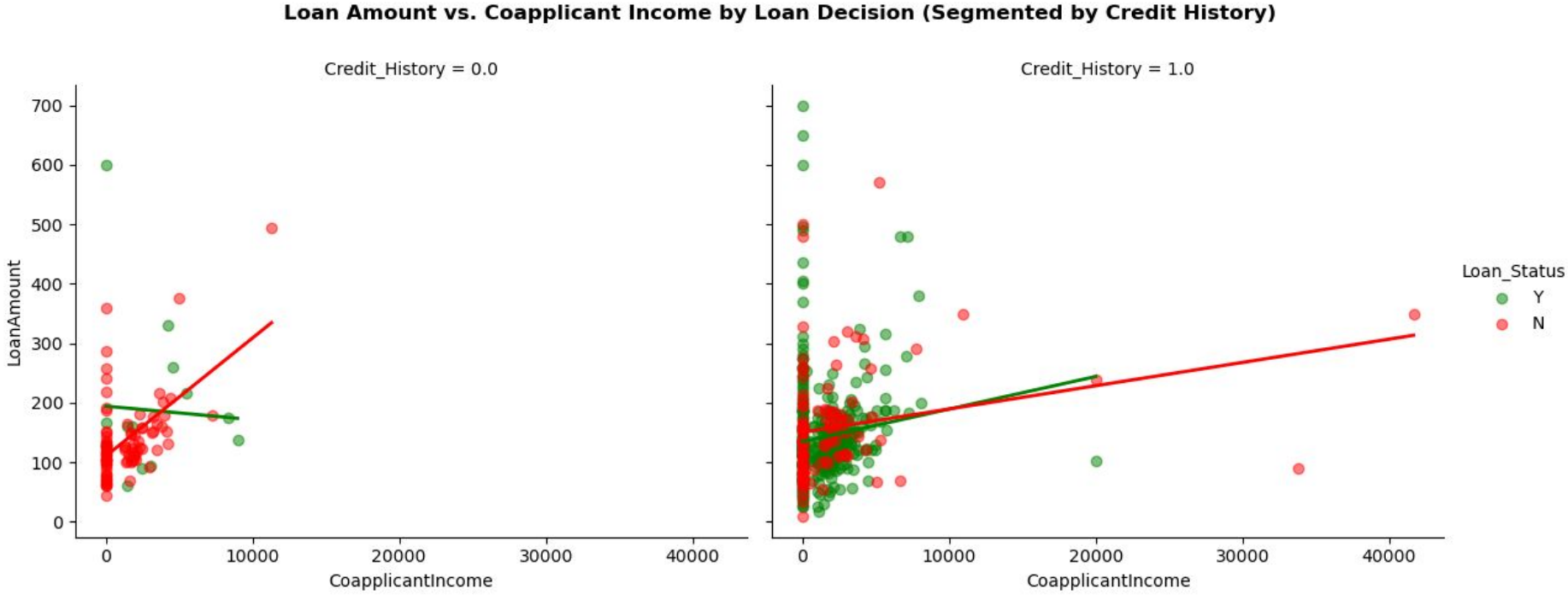
Higher Loan Amount requests are more associated with rejected loans.

Loan Approval Rate Analysis



- ✔ Applicant income is a strong factor in determining loan amount, while co-applicant income has a smaller effect.
- ✔ Loan terms are largely independent of applicant income and loan amount.
- ✔ Having a co-applicant may be a compensatory strategy for lower-income applicants, but it doesn't strongly influence loan amounts.

Loan Approval Rate Analysis

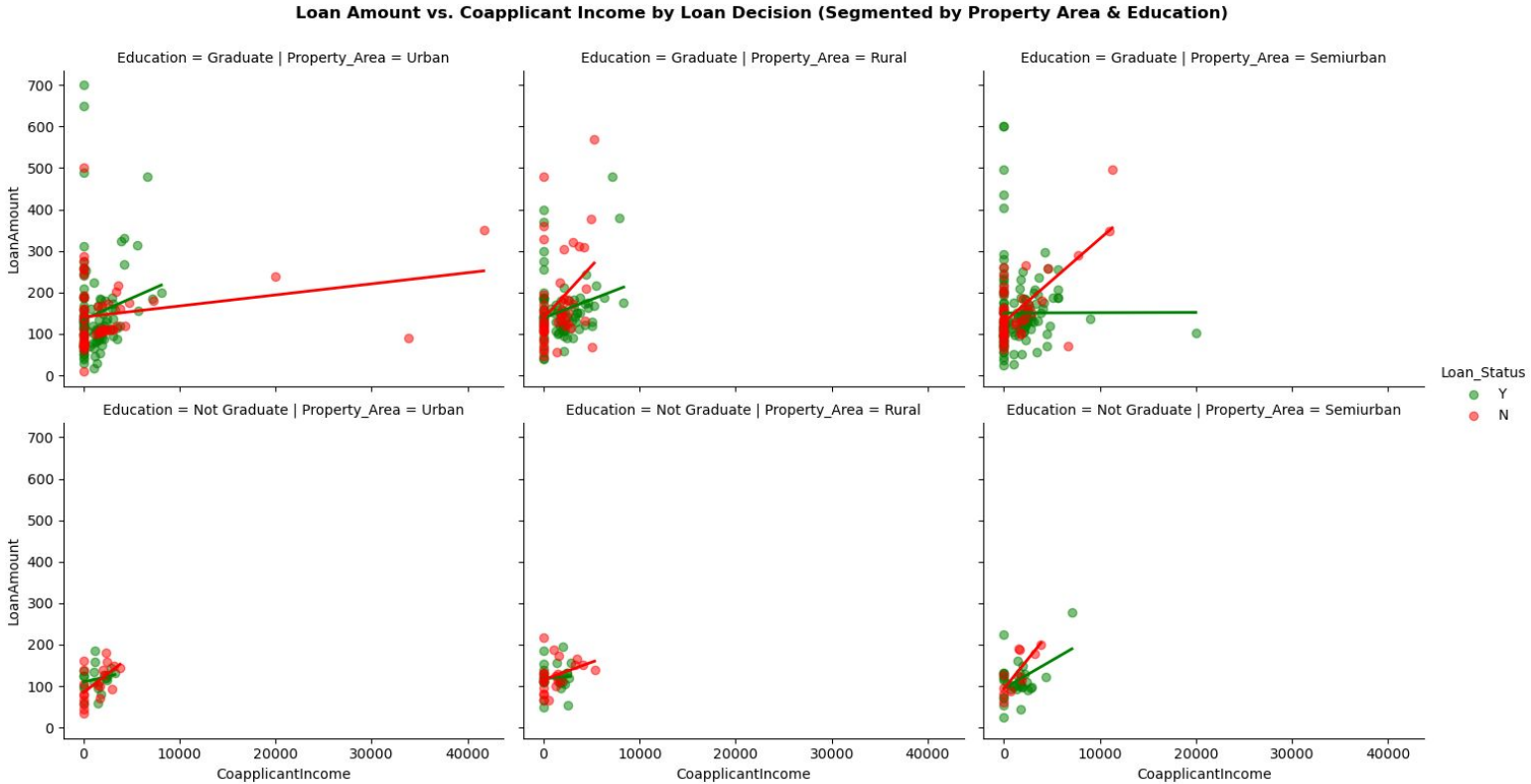


✔ Credit history is a game changer—with a strong credit history, a co-applicant's income significantly boosts approval chances.

✔ Without credit history, approvals remain low even for high-earning co-applicants.

✔ Loan approval likelihood improves when a co-applicant is present, especially for higher loan amounts.

Loan Approval Rate Analysis

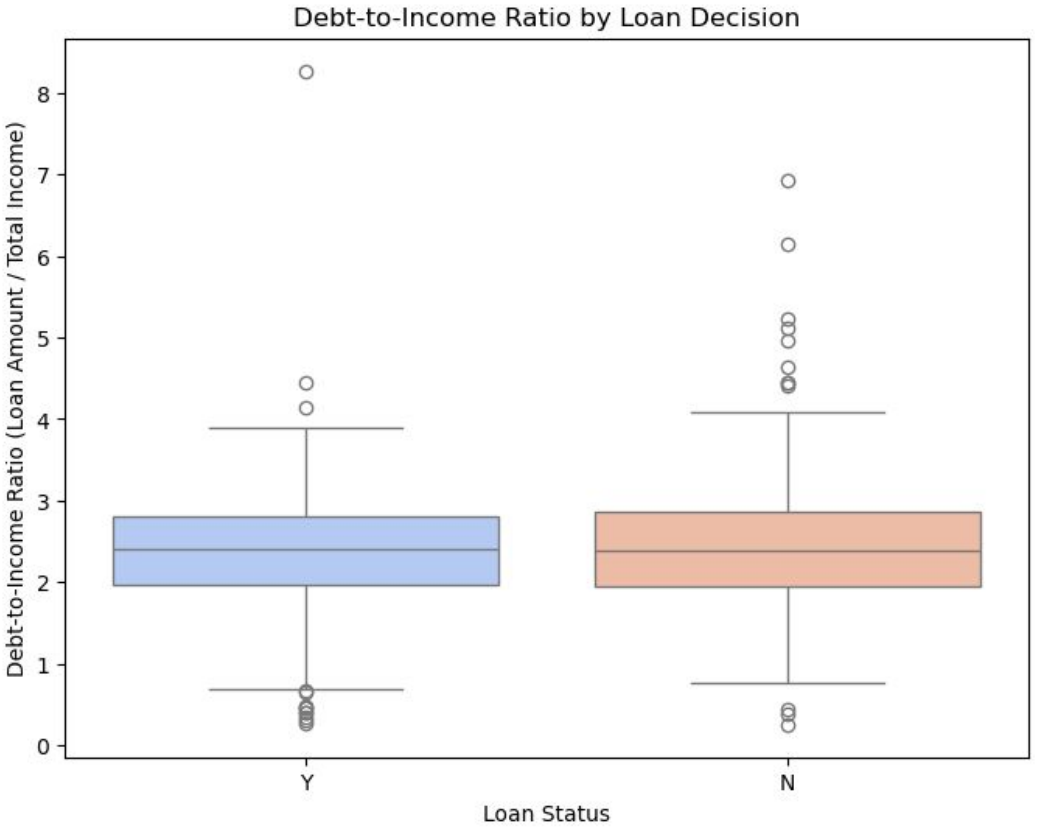


✔ Graduates in Urban and Semi Urban areas have the best approval chances, especially with a co-applicant.

✔ Non-graduates in Rural areas struggle to get approved, even with additional co-applicant income.

✔ Loan approvals are more flexible in Urban areas, where banks consider co-applicant income more seriously.

Loan Approval Rate Analysis

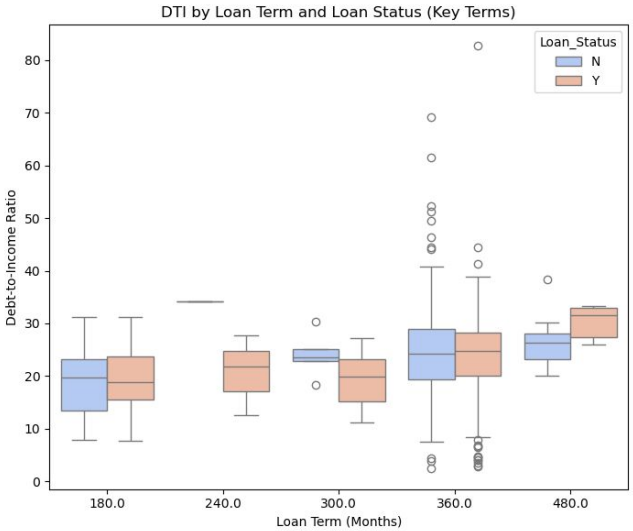


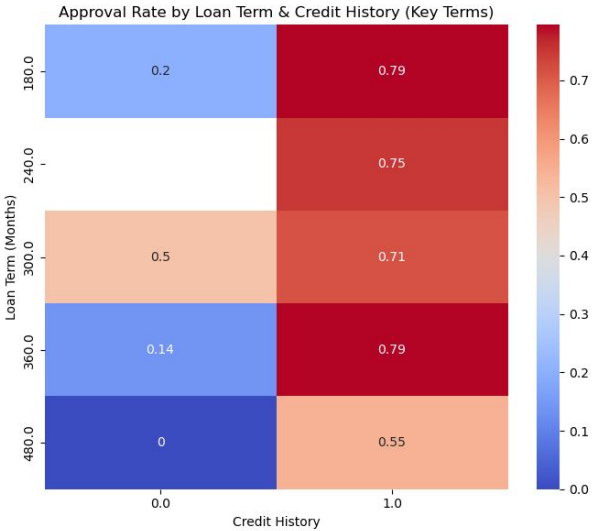
✔ Keeping Debt-to-income low increases the chances of loan approval.

✔ Banks set a threshold for how much debt an applicant can take based on their income.

✔ High-income applicants can still face rejections if their requested loan is disproportionately high.

Loan Approval Rate Analysis



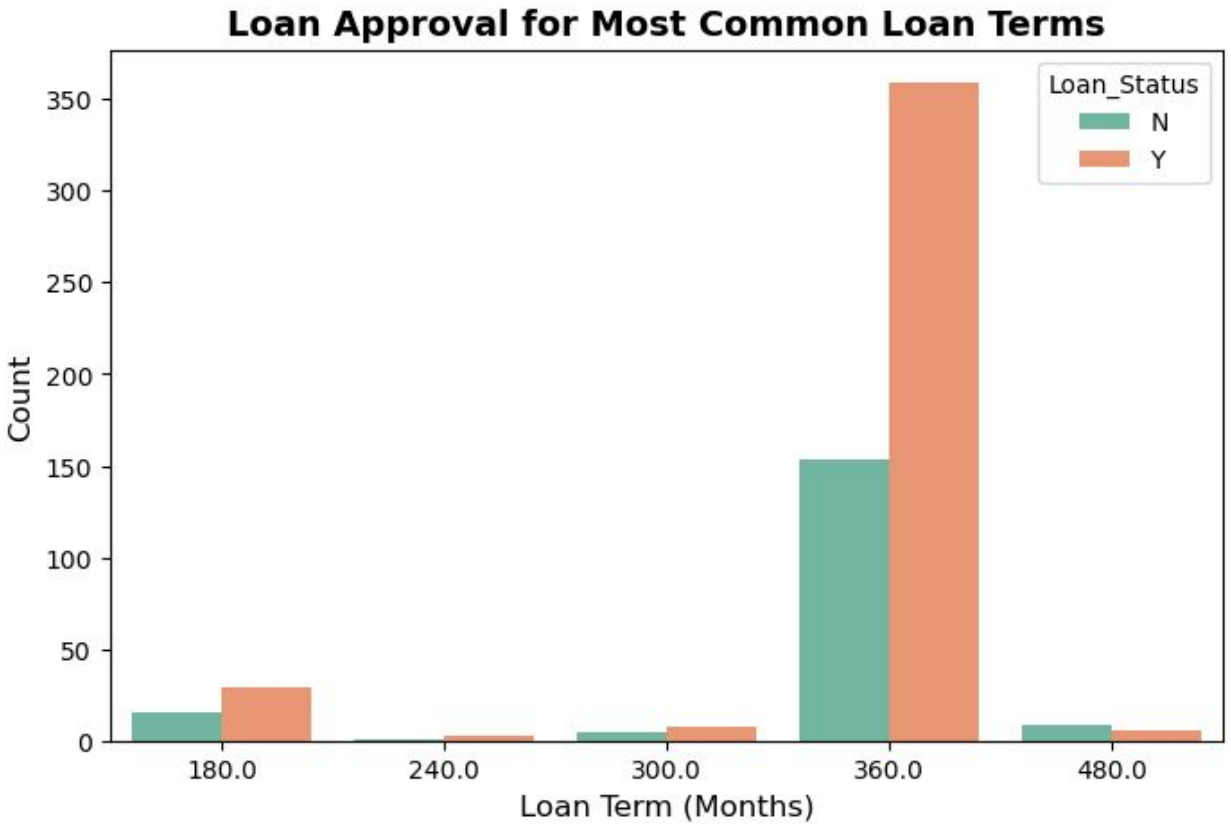


✔ Banks are more risk-averse in Rural areas—higher risk scores mean higher rejection rates.

✔ Urban areas allow for a broader range of risk acceptance, increasing approval flexibility.

✔ Longer loan terms (30 years) are preferred for higher-risk loans, especially in less developed areas.

Loan Approval Rate Analysis

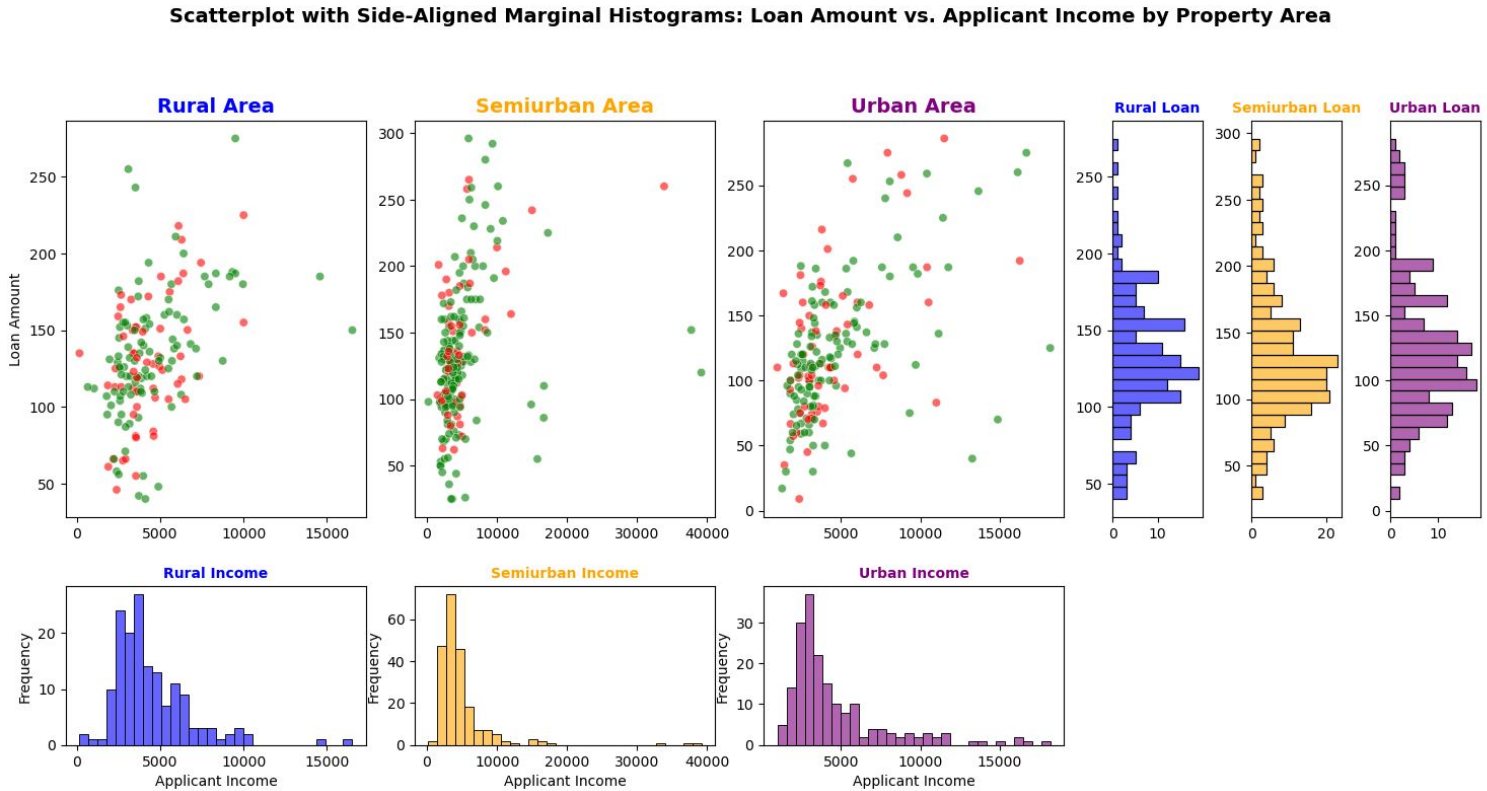


✔ Banks prefer 30-year repayment plans for loan approvals.

✔ Shorter loan terms are riskier due to high monthly payments.

✔ Extreme loan terms (too short or too long) see higher rejection rates.

Loan Approval Rate Analysis



- ✔ Urban Areas show higher applicant incomes and wider loan amount spreads.
- ✔ Semiurban loans are balanced across income and loan amount ranges.
- ✔ Rural loans remain lower in income and amount, with a tighter spread.

Loan Approval Rate Summary of Key Insights

- Applicants with no credit history have very low approval chances, even if they have high incomes.
- Gender: Slightly higher approval rate for male applicants.
- Marital Status: Married applicants have a higher approval rate than unmarried ones.
- Education: Graduates receive more approvals than non-graduates.
- Self-Employment: Self-employed applicants face slightly lower approval rates, likely due to income stability concerns.
- Higher co-applicant income improves approval rates.
- Loans with high amounts and no co-applicant income have higher rejection rates.
- Dual-income applicants are preferred, especially for high loan amounts.
- Semi-urban areas have the highest approval rates.
- Non-graduates in rural areas have higher rejection rates.
- Education level matters more in rural areas than in urban settings.
- 360-month (30-year) loan terms are most common and have the highest approval rates.
- Shorter loan terms (180 months or less) face more rejections.
- Extremely long (480-month) or short (12-month) terms also have high rejection rates.
- Higher DTI leads to more rejections.Banks favor applicants with a manageable debt load compared to their income.
- Rural areas have higher rejection rates for high-risk applicants.Urban and Semi-urban areas show more flexibility in approval.

Loan Approval Prediction Model

The goal of this project is to build and compare multiple machine learning models to predict loan approval (`Loan_Status`) based on applicants' financial and demographic data.

Data Preprocessing

Before building machine learning models, we preprocess and do feature engineering:

- **Debt-to-Income Ratio (DTI): Loan Amount / (Applicant + Co-Applicant income) *100**
  This feature helps to assess how much of an applicant’s income goes toward loan repayment.
- **One-Hot Encoding:**
  Categorical variables like `Gender`, `Married`, `Education`, `Self_Employed`, and `Property_Area` are converted into numeric format using one-hot encoding.
- **Defining Features and Target Variable:**
  - The `Loan_Status` column (Yes/No) is converted into binary values (`1` for approval, `0` for rejection).
  - The selected features include financial indicators (`ApplicantIncome`, `LoanAmount`, `Credit_History`, etc.) and demographic factors (`Gender`, `Education`, etc.).

Loan Approval Prediction Model

Model Selection & Training

To identify the best model for loan approval prediction, we train four different machine learning models:

1. Logistic Regression:
    - A probabilistic model ideal for binary classification.
    - Uses a logistic function to determine the probability of loan approval.
    - Standardized input features for optimal performance.
2. Random Forest Classifier:
    - An ensemble learning method using multiple decision trees.
    - Handles feature interactions well and is robust to overfitting.
3. XGBoost Classifier:
    - A boosting-based algorithm that improves weak classifiers iteratively.
    - Efficient in handling imbalanced datasets.
4. Support Vector Machine (SVM):
    - Uses hyperplanes to separate approved and rejected applications.
    - Requires feature scaling for better performance

Loan Approval Prediction Model

Model Evaluation & Cross-Validation

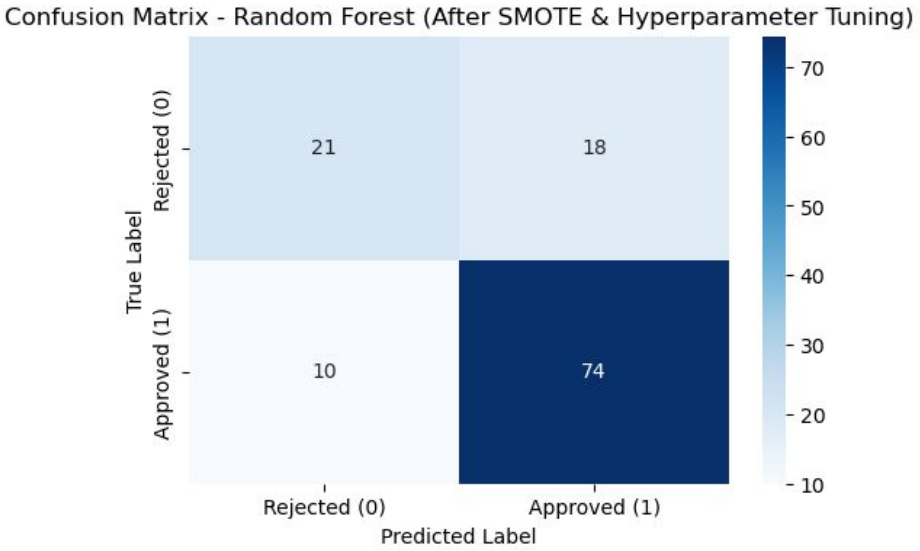
To ensure reliable model performance, we use Stratified K-Fold Cross-Validation (with 5 folds) for evaluation:

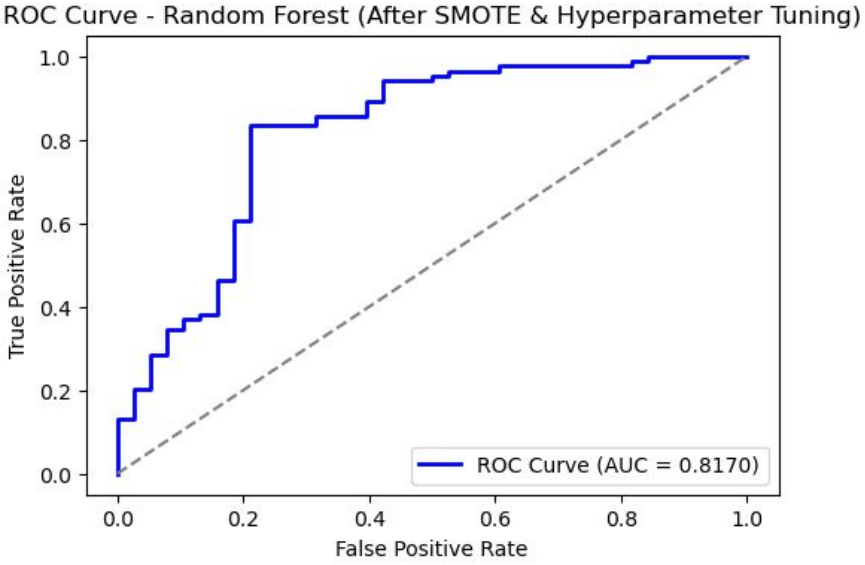
- Why Stratified K-Fold?
  It maintains the proportion of approved and rejected loans across all folds, ensuring balanced evaluation.
- Performance Metrics Used:
  - Accuracy: Measures how many total predictions were correct.
  - F1 Score: Balances precision and recall for imbalanced data.
  - ROC AUC Score: Evaluates how well the model distinguishes between approved and rejected applications.

Each model is trained and tested across multiple folds, and the average performance scores are recorded.

Model	Accuracy	F1 Score	ROC AUC
Logistic Regression	**75.085966**	**82.133798**	**75.107081**
Random Forest	**78.505931**	**85.534529**	**78.752746**
XGBoost	**75.409836**	**82.480748**	**76.476746**
SVM	**76.062908**	**83.377264**	**75.396769**

Loan Approval Prediction Model



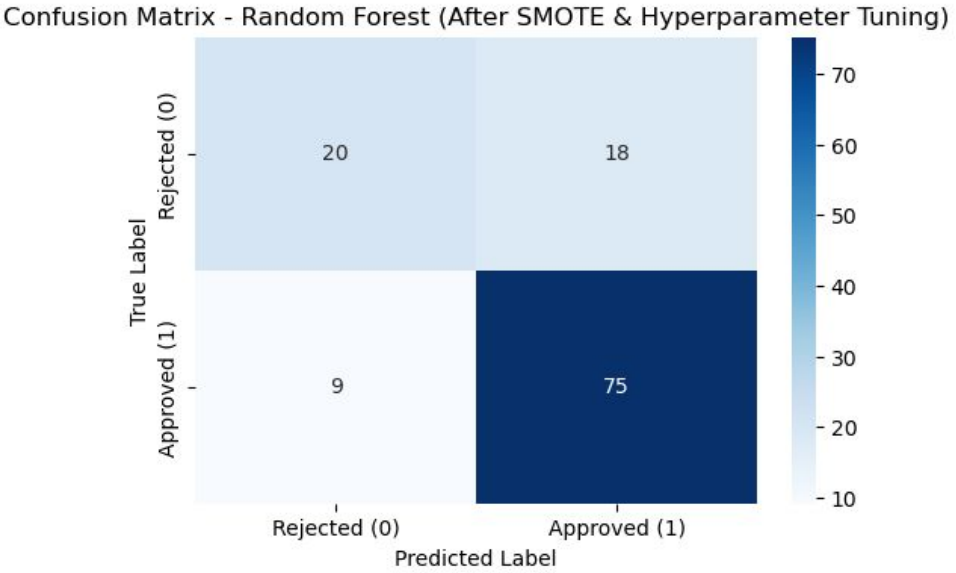


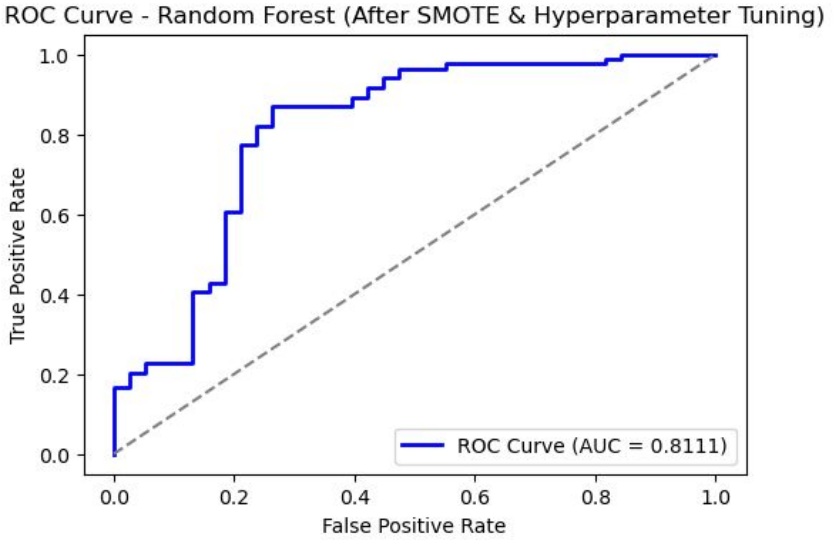
Loan Approval Prediction Model

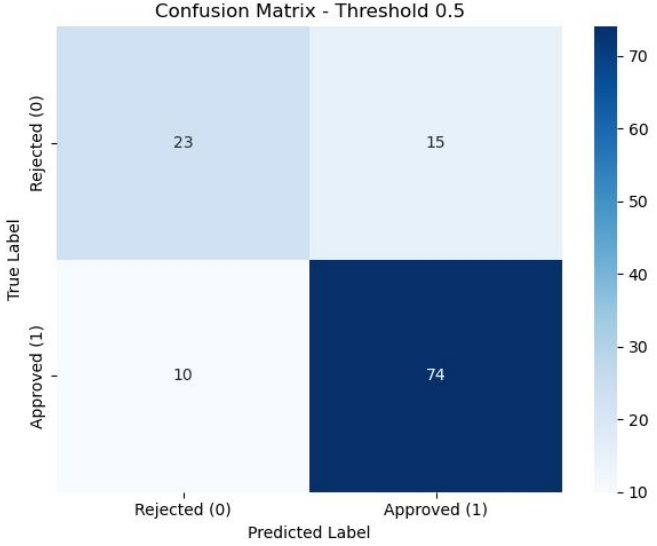
True Positives (TP): Correctly predicted loan approvals.
True Negatives (TN): Correctly predicted loan rejections.
False Positives (FP): Incorrectly predicted approvals (bad loans approved).
False Negatives (FN): Incorrectly predicted rejections (good loans rejected).

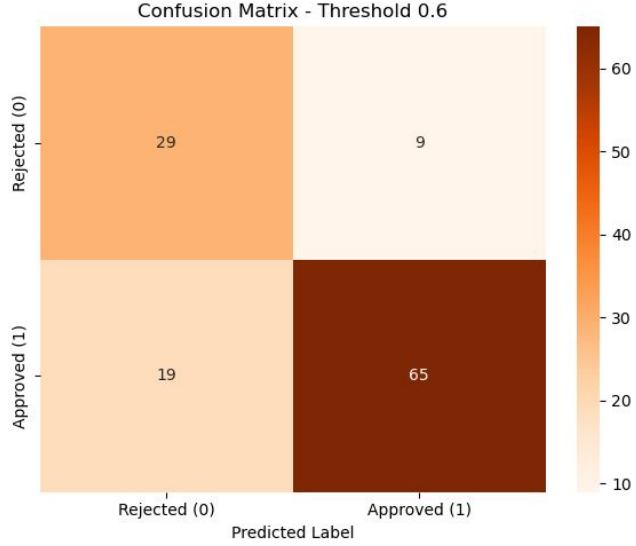
```
=== Threshold Analysis for False Positive Reduction ===

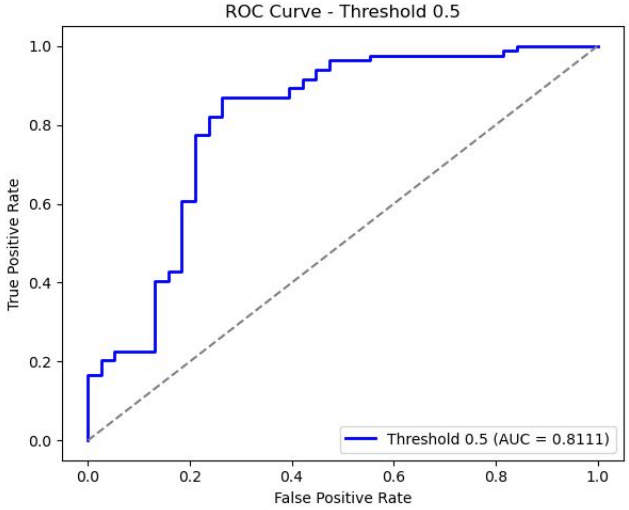
     Precision    Recall  F1 Score
0.4   0.822917  0.940476  0.877778
0.5   0.831461  0.880952  0.855491
0.6   0.878378  0.773810  0.822785
0.7   0.862745  0.523810  0.651852
```

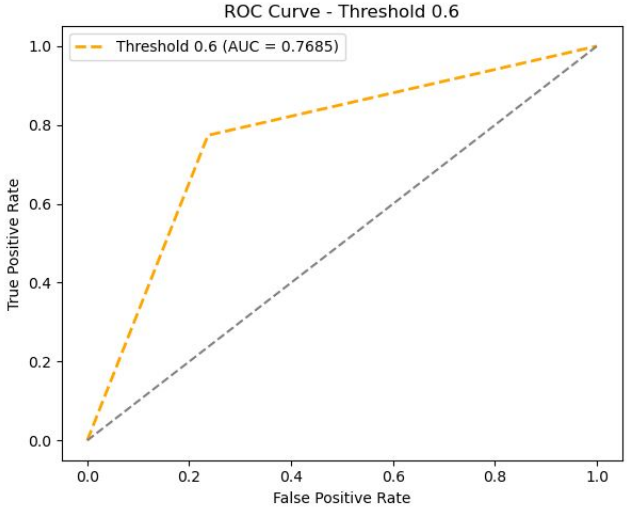


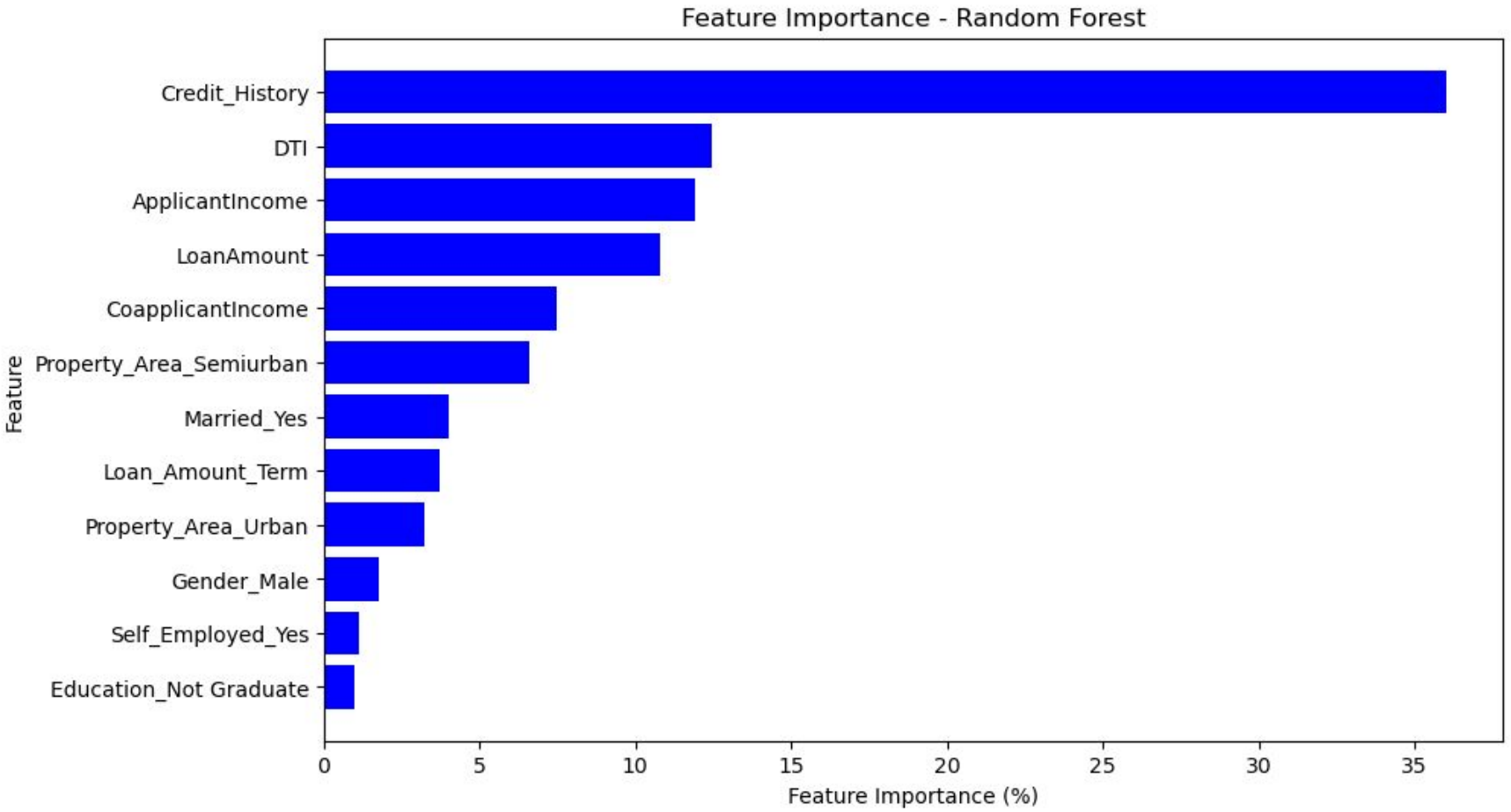



Confusion Matrix - Threshold 0.5
Rejected (0)
Approved (1)
True Label
23
15
10
74
Rejected (0)
Approved (1)
Predicted Label
70
60
50
40
30
20
10


Confusion Matrix - Threshold 0.6
Rejected (0)
Approved (1)
29
9
19
65
Rejected (0)
Approved (1)
Predicted Label
60
50
40
30
20
10


ROC Curve - Threshold 0.5
True Positive Rate
False Positive Rate
Threshold 0.5 (AUC = 0.8111)
0.0
0.2
0.4
0.6
0.8
1.0


ROC Curve - Threshold 0.6
True Positive Rate
False Positive Rate
Threshold 0.6 (AUC = 0.7685)
0.0
0.2
0.4
0.6
0.8
1.0


Feature Importance - Random Forest
Credit_History
DTI
ApplicantIncome
LoanAmount
CoapplicantIncome
Property_Area_Semiurban
Married_Yes
Loan_Amount_Term
Property_Area_Urban
Gender_Male
Self_Employed_Yes
Education_Not Graduate
Feature
0
5
10
15
20
25
30
35
Feature Importance (%)

Loan Approval Prediction Model Summary

- Random Forest is the best performance prediction in term of Accuracy, F1, AUC-ROC score
- To reduce the false positive prediction (bad loans approval), we try to increase the model threshold to 0.6 with the trade-off by rejection rate is increase
- The model feature important aligns with the loan approval analysis by credit history is the most important features on loan approval prediction.